

Deep Networks

MLP/FNN to Universal Function Approximation

Introduction to Deep Learning · Class 9

10 August

Outline

The Multilayer Perceptron, Formally

Universal Approximation, Revisited

How Wide Must a Network Be?

What This Settles

The Multilayer Perceptron, Formally

Notation used throughout this course

$\mathbf{x} \in \mathbb{R}^D$	input / features	$\phi_j(\mathbf{x})$	fixed basis function
$y, \hat{y}$	target, model prediction	$\mathcal{K} \subset \mathbb{R}^D$	a compact set
$\mathbf{z}^{(l)} = h(\mathbf{a}^{(l)})$	units of hidden layer l	$C(\mathcal{K})$	continuous functions on $\mathcal{K}$
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$	weights and biases	$\ f\ _\infty$	$\sup_{\mathbf{x} \in \mathcal{K}} f(\mathbf{x}) $
L, M	depth, width	ε	target accuracy
W	total number of parameters	$\mathcal{N}_{L,M}$	functions a network can compute

Symbols follow **Bishop & Bishop (2024)**. W counts every weight and bias; $\mathcal{N}_{L,M}$ is the set of functions a network of that shape can compute.

Where Class 7 left us

- Composition **multiplies** linear pieces; width only **adds** them
- Universal approximation holds in D dimensions for any non-polynomial activation, via ridge functions
- ReLU networks compute **exactly** the continuous piecewise-linear functions — shallow or deep
- The sawtooth needs $O(L)$ units deep and $2^L - 1$ shallow

All of that was about **what** a network can represent. None of it says **how much network** it takes.

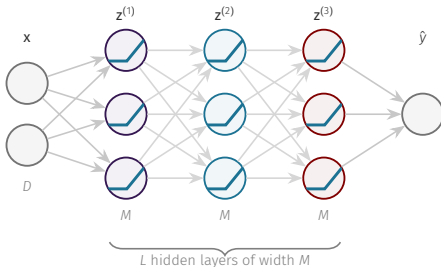
Three questions, and where they are answered

	Question	Answered by	
1	Can one hidden layer reach any function at all?	Cybenko, Hornik, Leshno	§2
2	How wide must a network be before it works at all?	Lu, Pu, Wang, Hu & Wang	§3
3	How wide must a network be before it works at all?	Lu, Pu, Wang, Hu & Wang	§3

A fourth question — how the budget should be *split* between depth and width — is the whole of **Class 10**.

The multilayer perceptron

Before we can count anything, we need to fix exactly what we are counting.



$$\mathbf{z}^{(0)} = \mathbf{x}, \quad \mathbf{z}^{(l)} = h(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}), \quad \hat{\mathbf{y}} = \mathbf{W}^{(L+1)}\mathbf{z}^{(L)} + \mathbf{b}^{(L+1)}$$

Write $\mathcal{N}_{L,M}$ for the set of functions this computes as the weights range over $\mathbb{R}^W$, with

$$W = M(D+1) + (L-1)M(M+1) + (M+1) \text{ parameters.}$$

Why not just use fixed basis functions?

Before calling learned features progress, see what goes wrong without them.

Tile the input space, one basis function per cell (B&B, 2024, §6.1):

$$\hat{y}(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x}), \quad \phi_j \text{ centred on cell } j$$

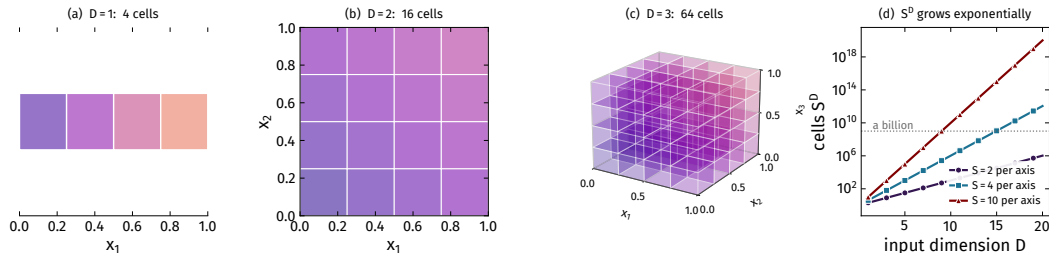
Proposition 1 (The grid blows up)

Partition each of the D axes into S intervals. The tensor-product grid then has S^D cells, so the model has S^D parameters — exponential in the input dimension.

Proof. A cell is one interval chosen per axis, so there are $|\{1, \dots, S\}^D| = S^D$. □

With $S = 10$ and a 32×32 image ($D = 1024$): 10^{1024} cells.

The curse of dimensionality, drawn

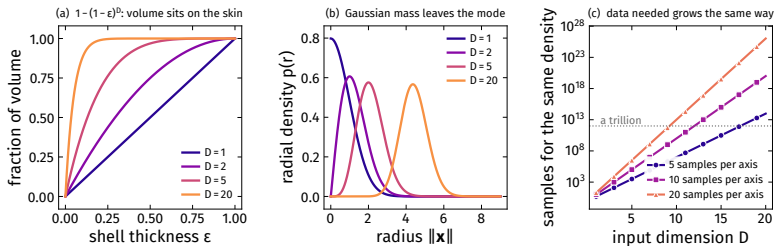


(a–c) a grid with $S = 4$ cells per axis in one, two and three dimensions: 4, 16, 64 cells. (d) the count S^D against dimension, at three resolutions.

Every cell needs its own parameter — and its own data to fit it.

Two more faces of the same problem

High-dimensional space is strange in two further ways that matter later.



(a) the volume within ϵ of a unit ball's surface, $1 - (1 - \epsilon)^D$. (b) a Gaussian's mass migrates to a shell at radius $\approx \sqrt{D}$. (c) matching a sample density costs exponentially more data.

SHALLOW fixed ϕ_j : one parameter per cell, S^D of them

DEEP learned features: units *move* to where the data is

Universal Approximation, Revisited

What “universal” means

Networks are supposed to escape this. First, say precisely what that claim asserts.

Fix a compact $\mathcal{K} \subset \mathbb{R}^D$ and use the supremum norm $\|f\|_\infty = \sup_{\mathbf{x} \in \mathcal{K}} |f(\mathbf{x})|$.

Definition

A family $\mathcal{F} \subset C(\mathcal{K})$ is **universal** (dense) if for every $f \in C(\mathcal{K})$ and every $\varepsilon > 0$ there is $g \in \mathcal{F}$ with $\|f - g\|_\infty < \varepsilon$.

- Note what is quantified: $\mathcal{F}$ is the family with **unbounded** width, so “ g exists” may mean a network of astronomical size
- Density says nothing about **how fast** the error falls as the network grows — that needs a **rate**

The classical statement

With the definition in place, the theorem is short.

Theorem 2 (Universal approximation; Cybenko 1989, Hornik 1991, Leshno et al. 1993)

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. The family

$$\mathcal{N}_{1,\infty} = \left\{ \mathbf{x} \mapsto \sum_{j=1}^M w_j^{(2)} h(\mathbf{w}_j^{(1)\top} \mathbf{x} + w_{j0}^{(1)}) + w_0^{(2)} : M \in \mathbb{N} \right\}$$

is dense in $C(\mathcal{K})$ for every compact $\mathcal{K}$ if and only if h is not a polynomial.

- Proved in Class 7 by reduction to one dimension through **ridge functions**
- **One hidden layer is enough** — so universality alone can never be an argument for depth

How Wide Must a
Network Be?

Width as a threshold

We know the cost of accuracy. Now: how to spend the budget. Width behaves quite unlike depth.

Theorem 3 (Width-bounded universality; Lu, Pu, Wang, Hu & Wang, 2017)

*ReLU networks of width $D + 4$ and **arbitrary depth** are dense in $L^1(\mathbb{R}^D)$: for every Lebesgue-integrable f and every $\varepsilon > 0$ there is such a network with $\int |f - \hat{y}| \, d\mathbf{x} < \varepsilon$.*

- Four units more than the input dimension is **all** the width that is ever needed, provided the network may be deep
- Note the norm: density in L^1 , not uniform approximation

Width is a threshold: a small fixed margin above D is all that is ever needed. **Depth** is a resource.

What This Settles

What each theorem actually gives

Ten results in one lecture. What each one is really claiming:

Result	Gives	Does not give
Cybenko / Hornik / Leshno	density with one hidden layer, any non-polynomial h	any bound on the width

What none of them say

And the gap between all this and a working network:

- **Nothing about optimisation.** Every result is about the *existence* of weights; finding them is Classes 11–14
- **Nothing about generalisation.** The capacity bounds are worst-case over all data distributions
- **Architecture matters more than the counts suggest.** Convolution and attention change the function class itself

SHALLOW universal, with a vacuous width bound

DEEP the useful question is what the budget *buys*
— Class 10

Takeaways

1. Fixed basis functions need S^D parameters. Networks avoid this by **placing their features**, not tiling the space
2. One hidden layer is already universal, for any non-polynomial activation — so **universality alone never argues for depth**
3. A grid of fixed features costs S^D parameters; a network instead **places** its features where the data is
4. **Width is a threshold**: $D + 4$ always suffices given enough depth — width is something to clear, not to tune
5. Capacity counts both ways: regions grow exponentially in L , and so does the worst-case **VC bound**

Class 10: universality and rates say nothing about *how the budget is shaped*. Next: depth against width.

References

- C. M. Bishop & H. Bishop. *Deep Learning: Foundations and Concepts*. Springer, 2024. §6.1 (limitations of fixed basis functions, the curse of dimensionality), §6.2 (multilayer networks), §6.3 (deep networks).
- S. J. D. Prince. *Understanding Deep Learning*. MIT Press, 2023. Ch. 4 — the universal approximation theorem and its width version. The depth- and width-efficiency results from the same notes are Class 10.
- G. Cybenko (1989); K. Hornik (1991); M. Leshno, V. Lin, A. Pinkus & S. Schocken, *Neural Networks* 6:861 (1993).
- Z. Lu, H. Pu, F. Wang, Z. Hu & L. Wang, *NeurIPS* (2017) — width-bounded universality.
- Plots generated by `../B_Recreated_Figures/src/build_all.py`.
- This variant carries only results stated or cited in the two textbooks. For Barron (1993), DeVore et al. (1989), Yarotsky (2017) and the width lower bound, see variant A or B.

Questions?